

Big Data Programming

Python & Development Environment

Introduction to Python

Overview

Python 개요

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Installation

↓

VS Code 설치 및 실행

↓

Run Code 실행

↓

Variables & Expressions

목표: 직접 Python 환경을 구성하고 첫 번째 프로그램을 실행한다.

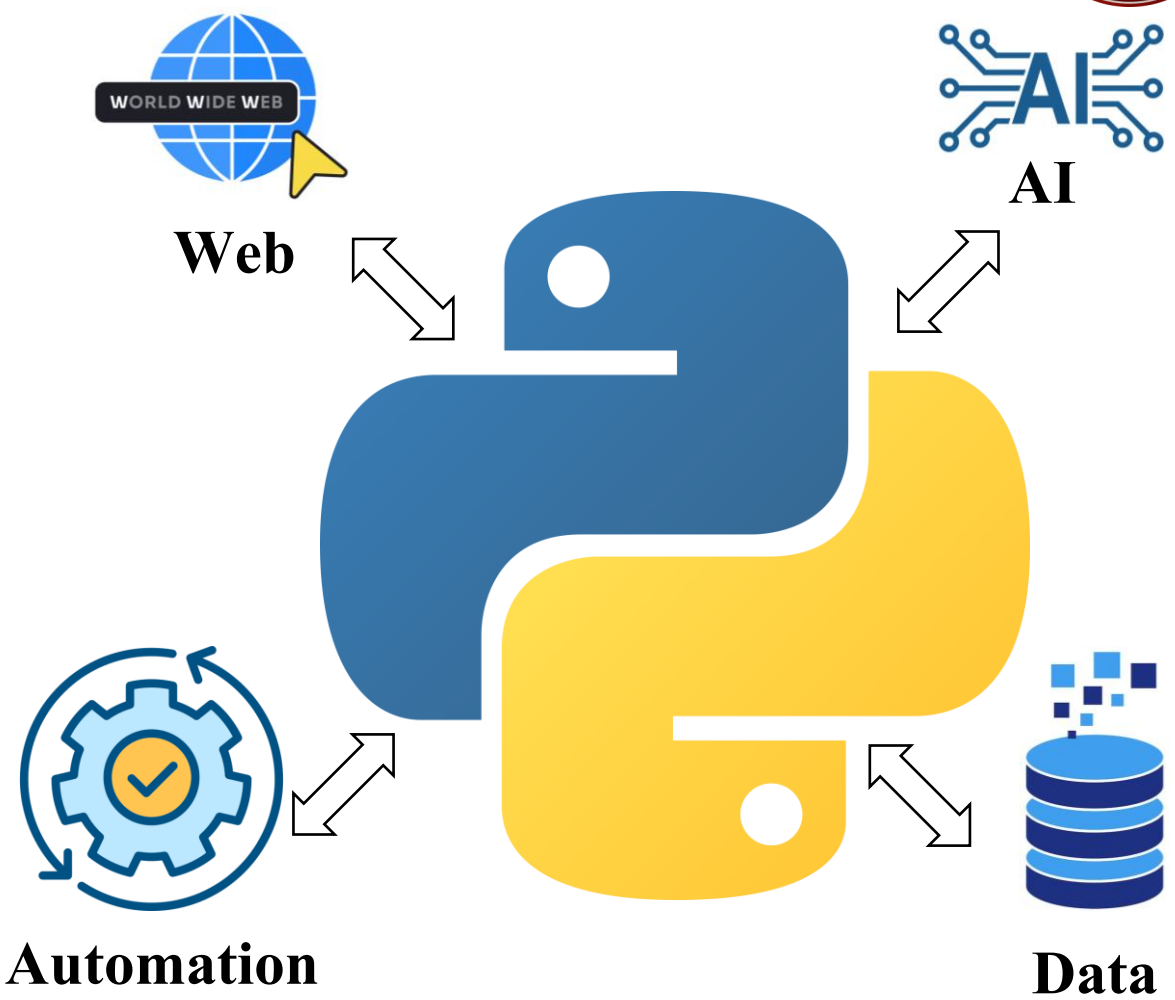
Introduction to Python

파이썬은 무엇인가? (What is Python?)

Python

- 1991년 Guido van Rossum이 공개
- 읽기 쉬운 문법
- 범용 프로그래밍 언어
- 다양한 분야에서 활용

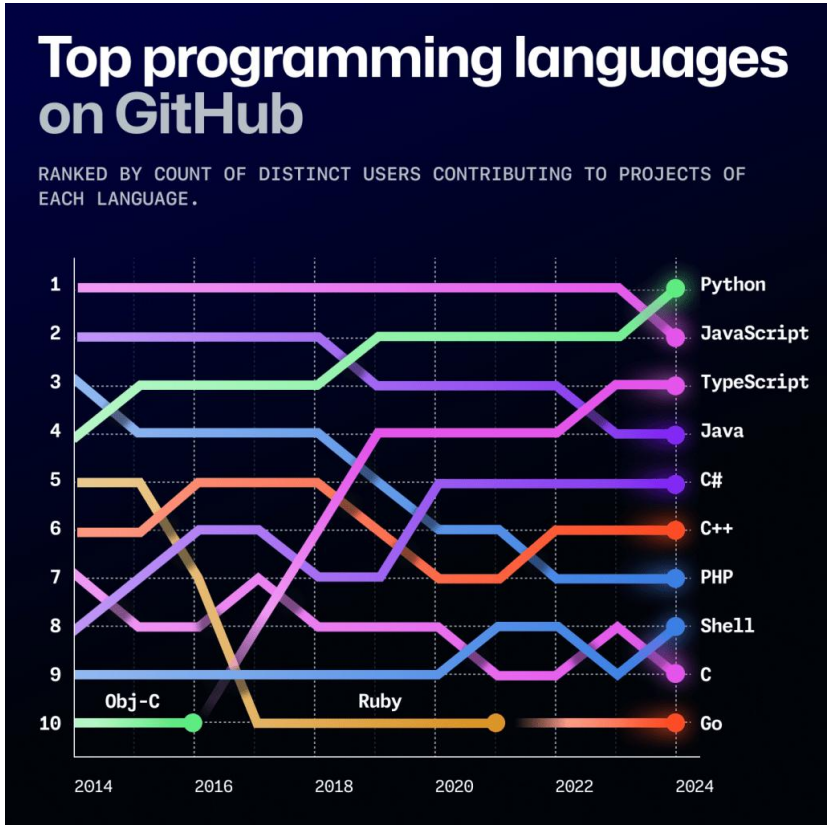




Python은 간결한 문법과 폭넓은 생태계를 가진 범용 프로그래밍 언어이다.

Introduction to Python

왜 Python을 배우는가? (Why Python?)



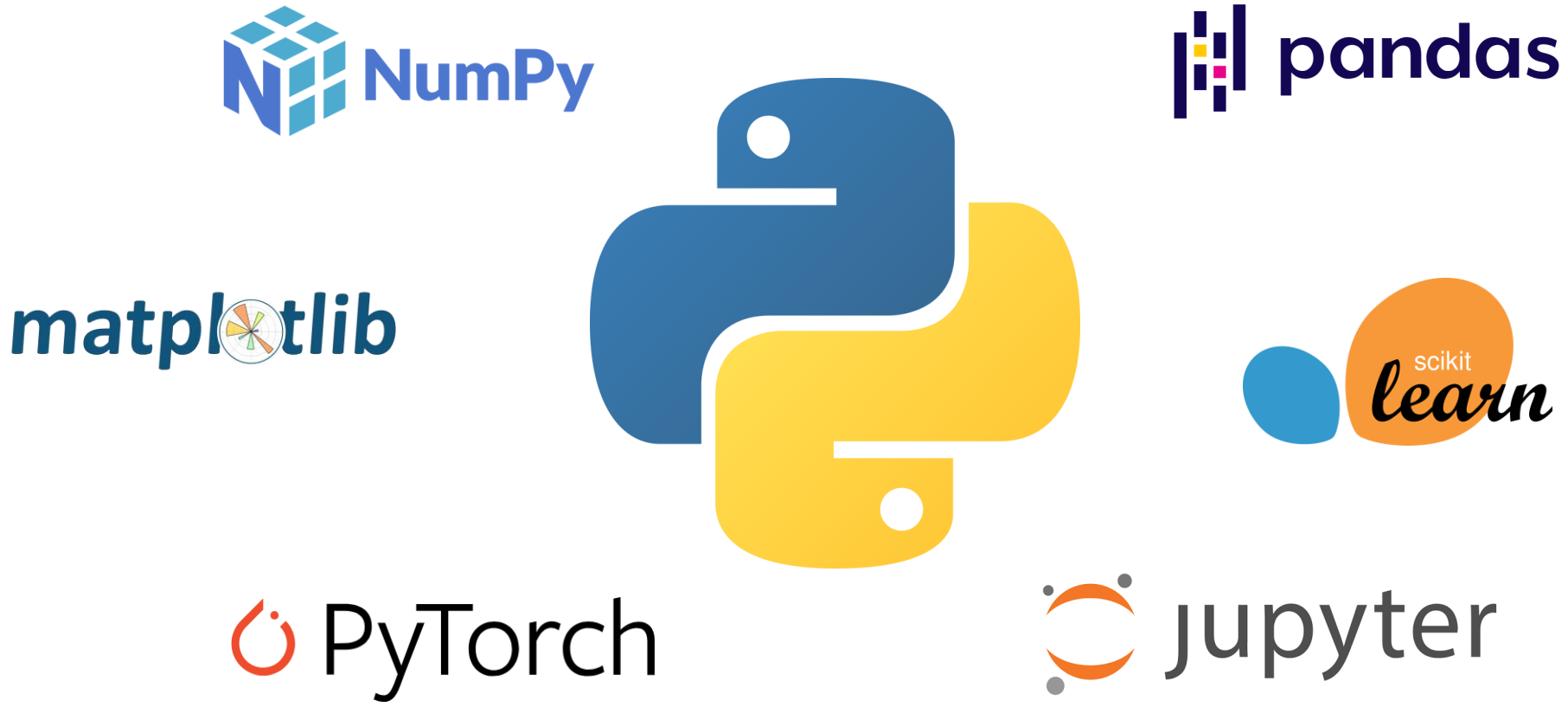
#1 on GitHub (2024)

- Artificial Intelligence
- Data Science
- Automation

Python은 다양한 분야에서 활용되며, 특히 AI와 데이터 사이언스의 핵심 프로그래밍 언어이다.

Introduction to Python

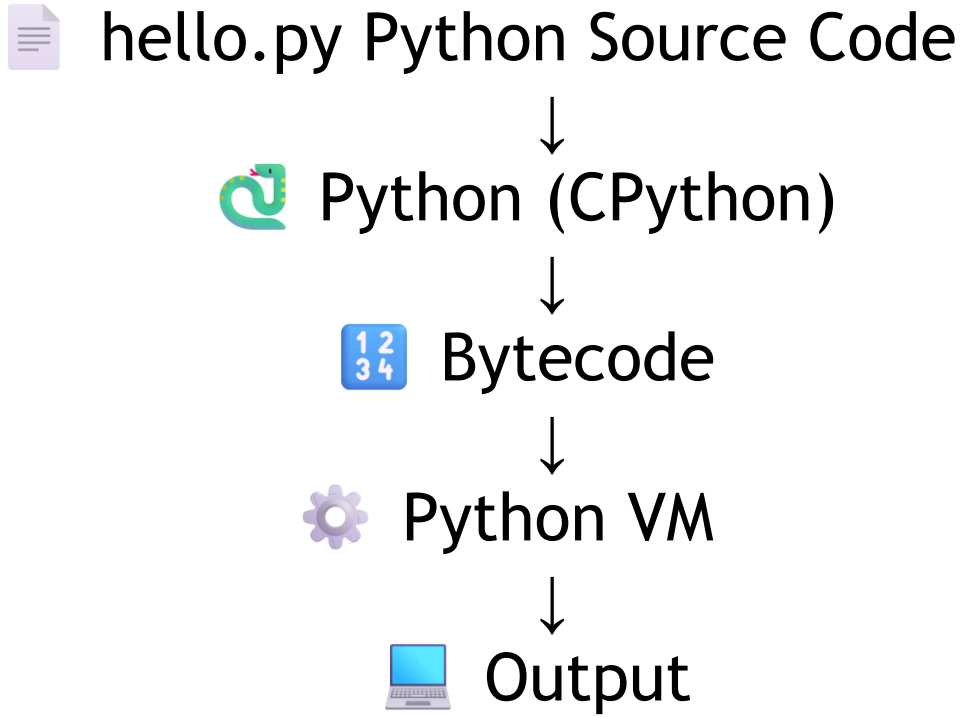
Python Ecosystem



하나의 언어 위에서 데이터 처리, 시각화, 머신러닝, 딥러닝까지 수행할 수 있다.

Introduction to Python

Python 코드는 어떻게 실행될까?



코드가 어떻게 내부적으로 실행 되는지 보다 먼저,
Python을 어떻게 사용 하는지부터 배운다.

Development Environment

Python을 사용하려면 무엇이 필요할까?

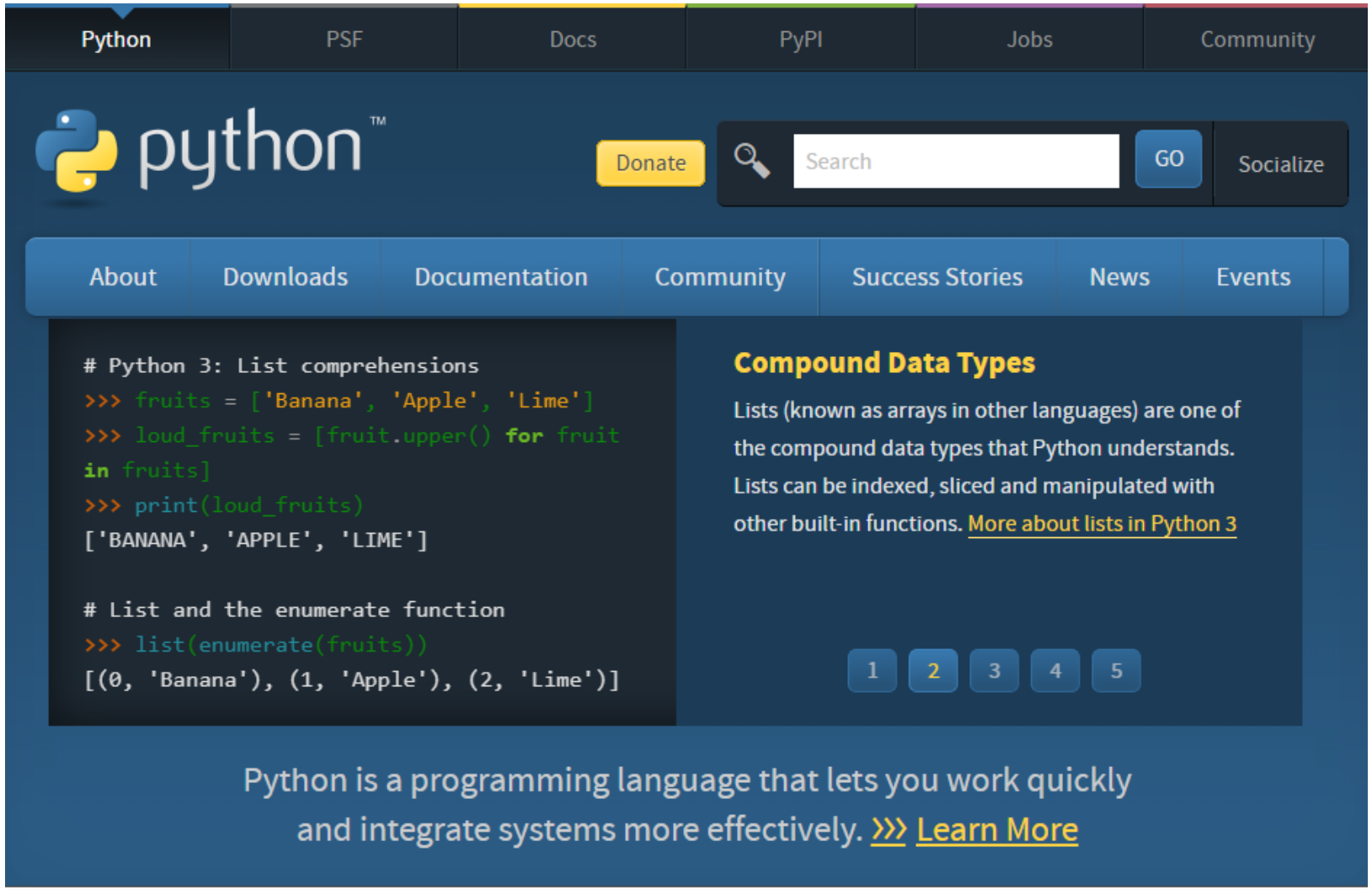


Writing → Running → Debugging

Python과 VS Code는 서로 다른 프로그램이며,
VS Code가 Python을 실행하도록 연결해야 한다.

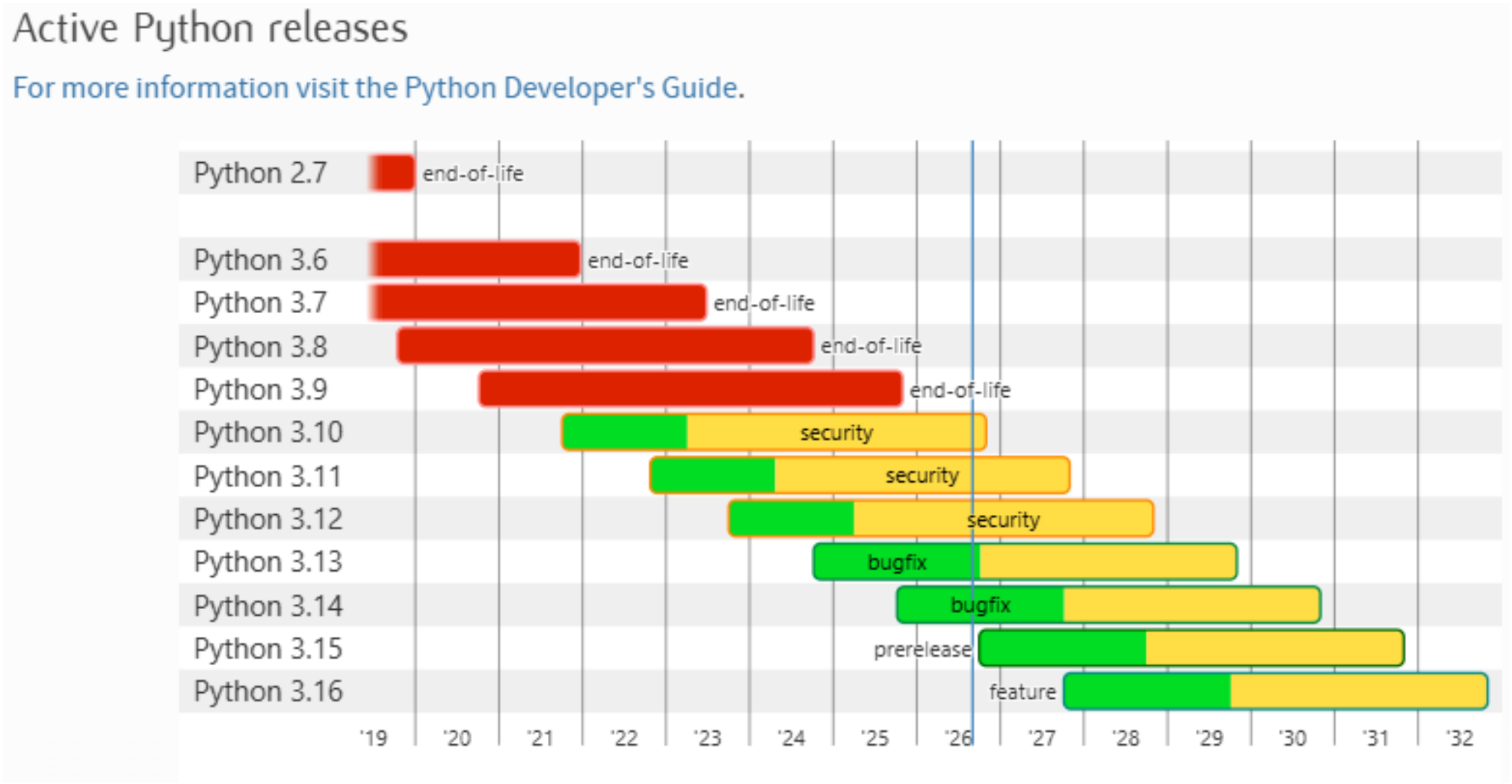
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Step 1. Python 설치하기 (https://www.python.org/)



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Looking for a specific release?

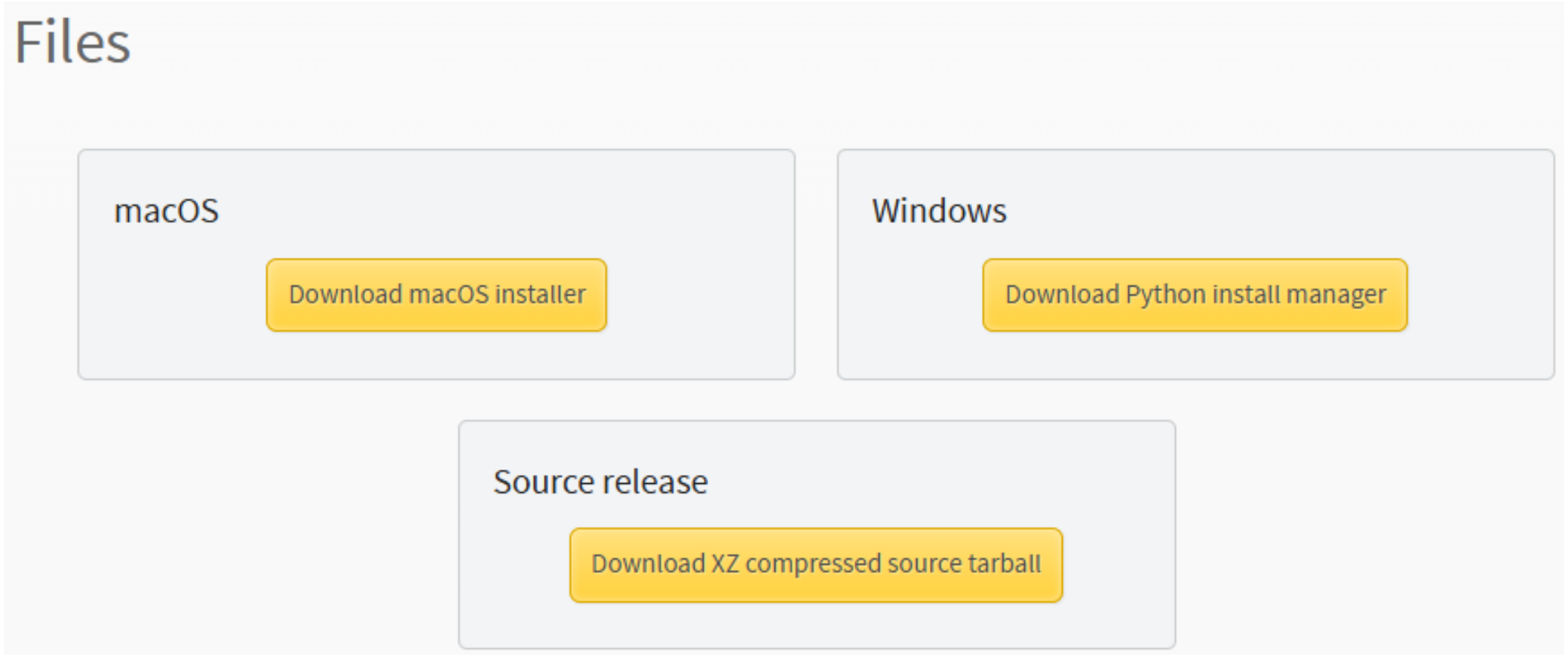
Python releases by version number:

Release version	Release date		Click for more
Python 3.14.7	Aug. 5, 2026	Download	Release notes
Python 3.14.6	June 10, 2026	Download	Release notes
Python 3.14.5	May 10, 2026	Download	Release notes
Python 3.14.4	April 7, 2026	Download	Release notes
Python 3.14.3	Feb. 3, 2026	Download	Release notes
Python 3.14.2	Dec. 5, 2025	Download	Release notes
Python 3.14.1	Dec. 2, 2025	Download	Release notes
Python 3.14.0	Oct. 7, 2025	Download	Release notes

View older releases

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Step 1. Python 설치하기 (https://www.python.org/)



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Files

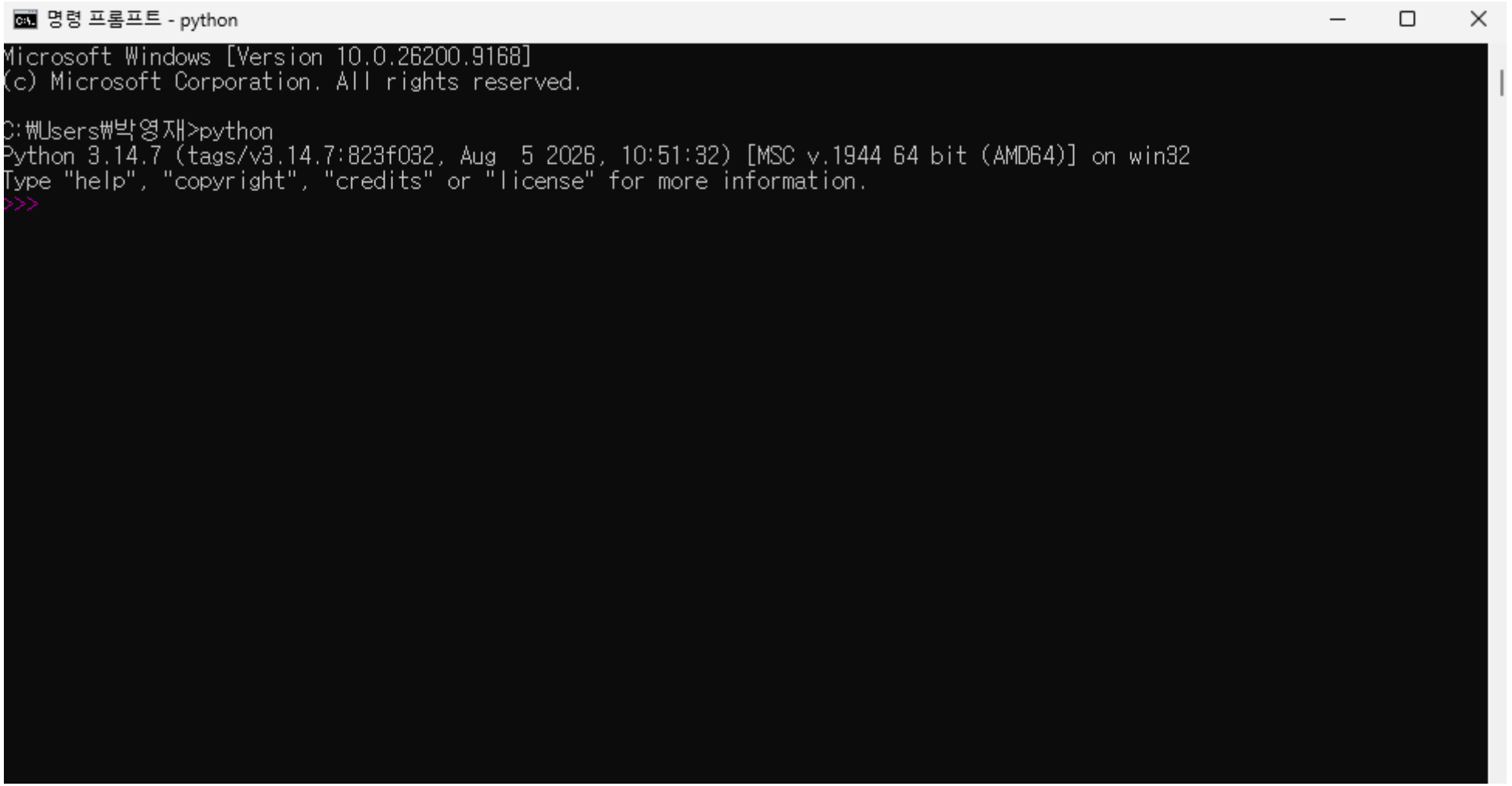
Windows

Download Installer (MSIX)

Version	Operating system	Description	File size	SHA-256 checksum
Installer (MSIX)	Windows	Bundles Python 3.14.6	44.9 MB	BDD1A4E0B485F674748B275095CC8E71 32BEB37A1AE91DB080ABB2661E3BADEC
MSI package	Windows	See documentation before use	9.4 MB	259AF5272C8F798786C1109B7AD287DA 519C58D43D7250EAD8ED20FA3277A511

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Python 설치 확인하기

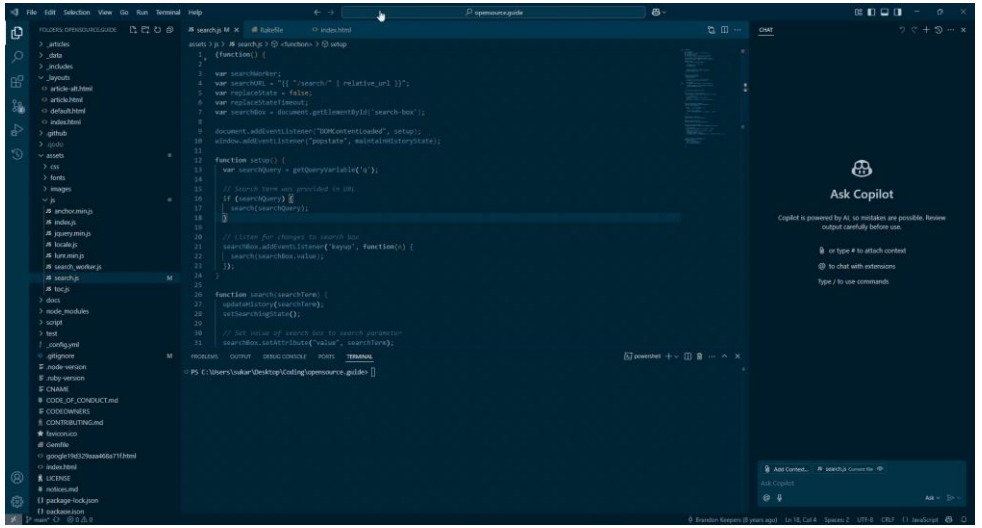


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Step 2. Visual Studio Code 설치하기 (https://code.visualstudio.com/)

VS Code

- 다양한 프로그래밍 언어를 지원하는 코드 편집기
- Python 확장 기능을 통해 작성·실행·디버깅 가능
- 가볍고 다양한 개발 도구를 쉽게 추가할 수 있음



이번 수업에서는 Python 코드를 VS Code에서 작성하고 실행한다.

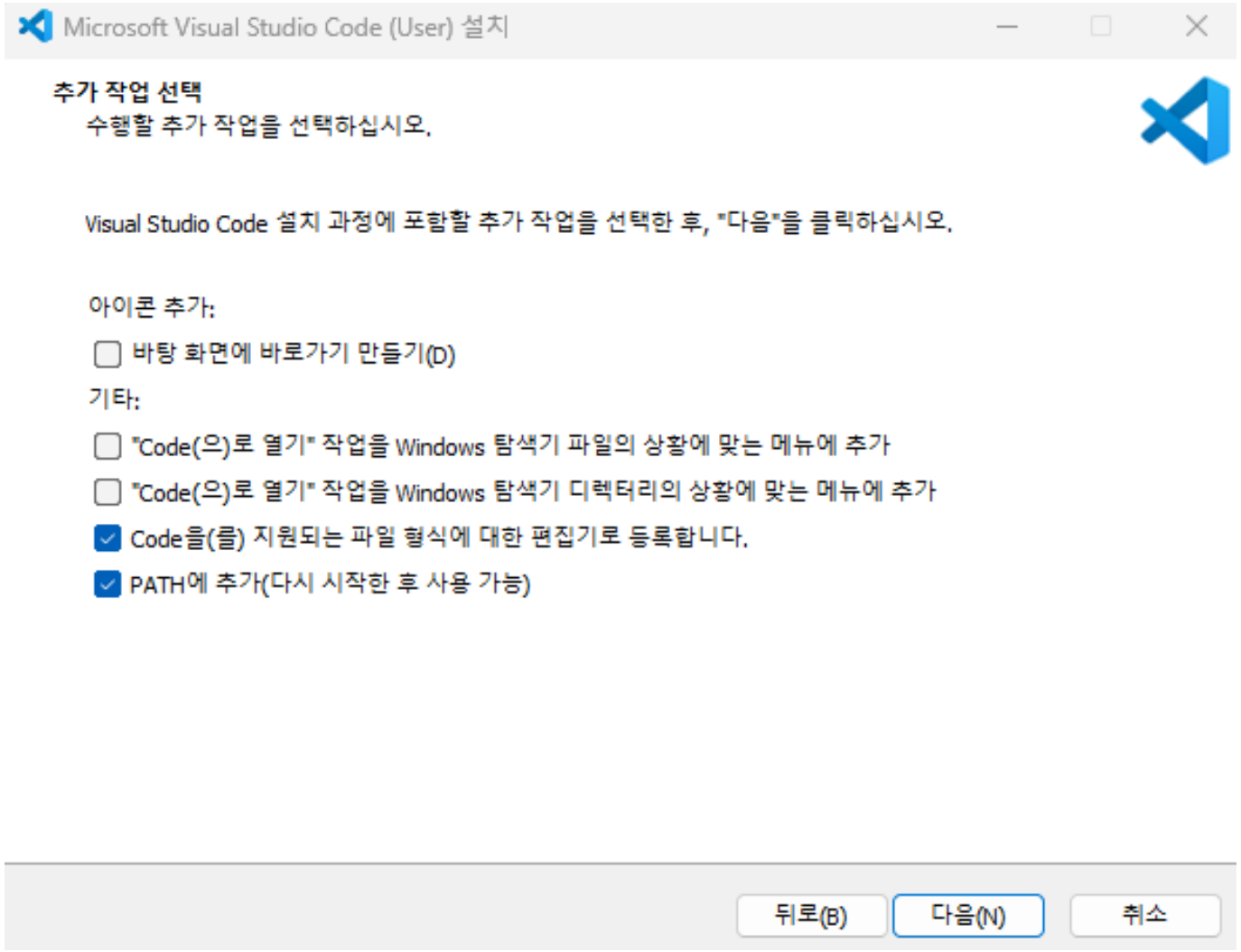
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Step 2. Visual Studio Code 설치하기 (https://code.visualstudio.com/)



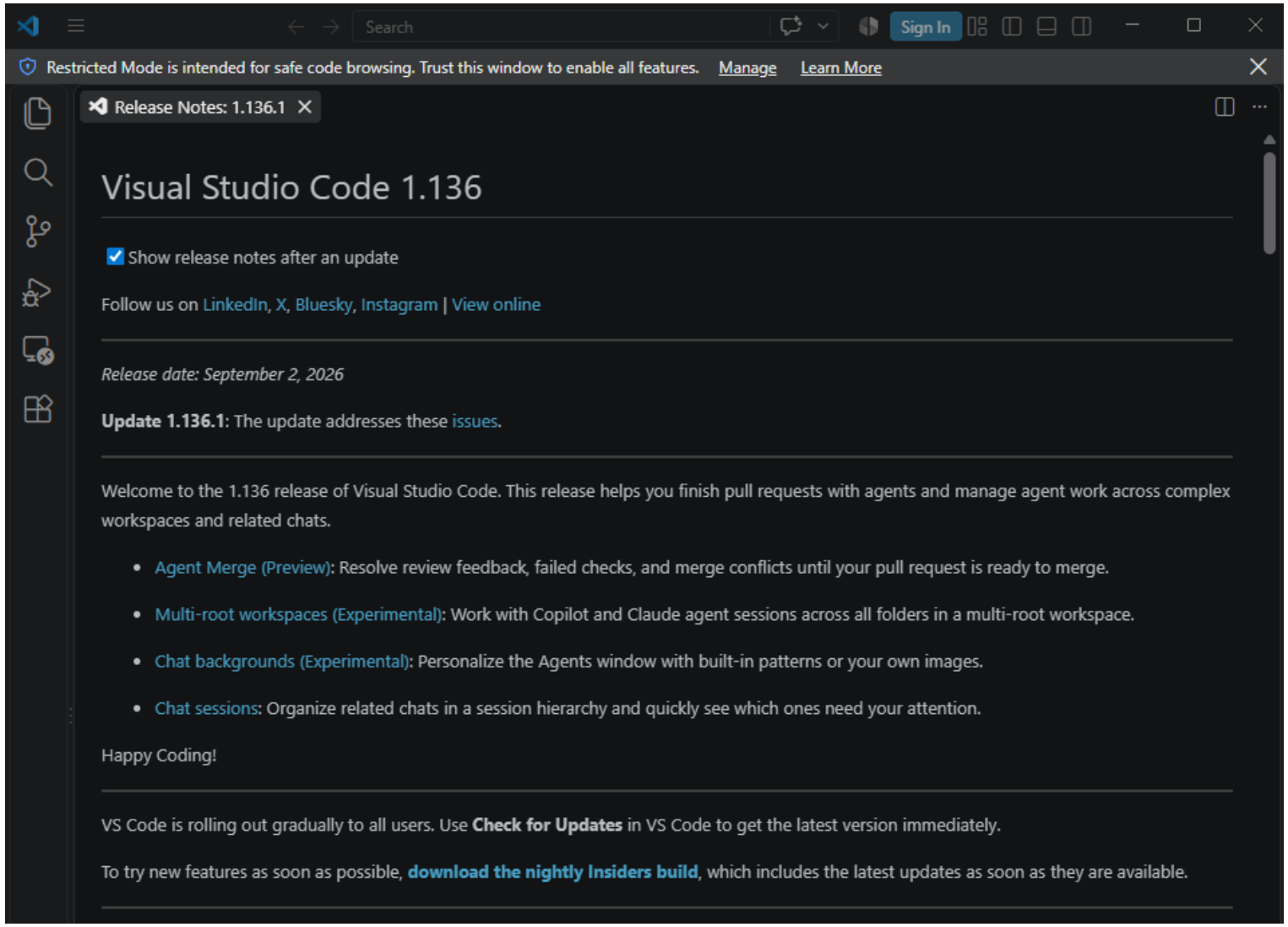
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Step 2. Visual Studio Code 설치하기 (https://code.visualstudio.com/)



Development Environment

Step 2. Visual Studio Code 설치하기 (https://code.visualstudio.com/)

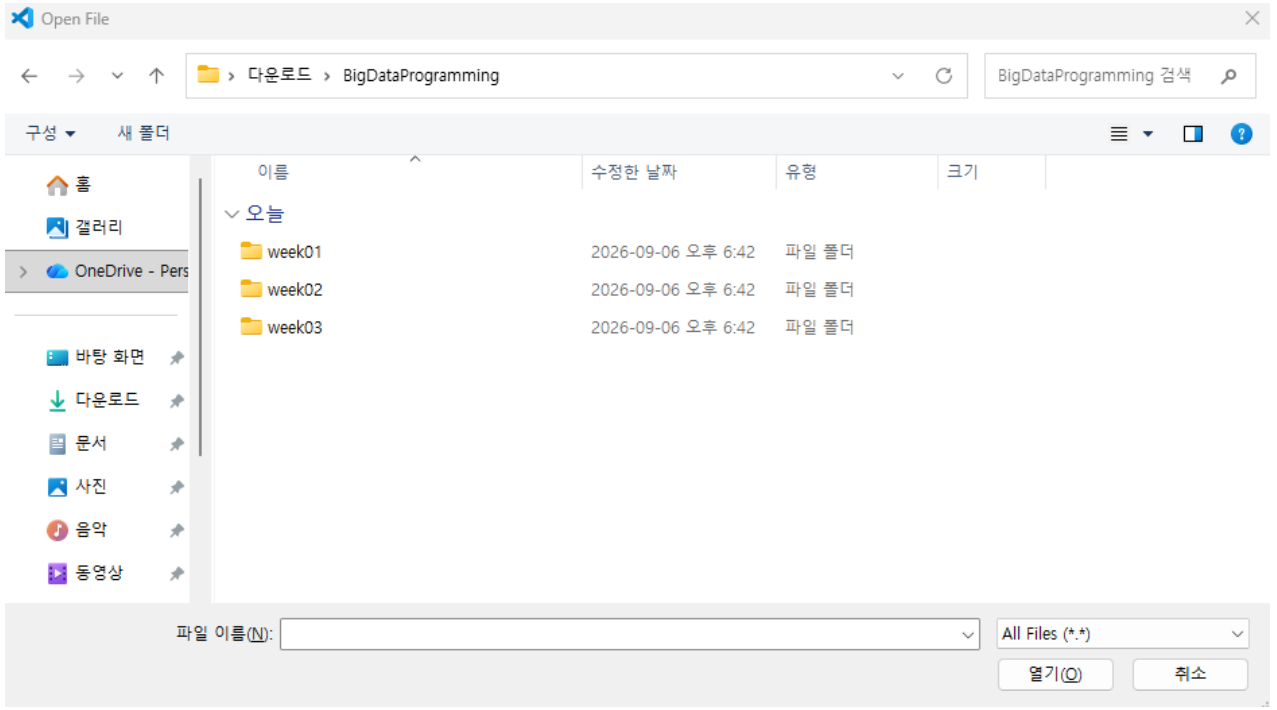


Development Environment

프로젝트 폴더 만들기

BigDataProgramming/

```
├── week01/
├── week02/
└── week03/
```



코드는 폴더에 정리해서 관리한다.

Development Environment

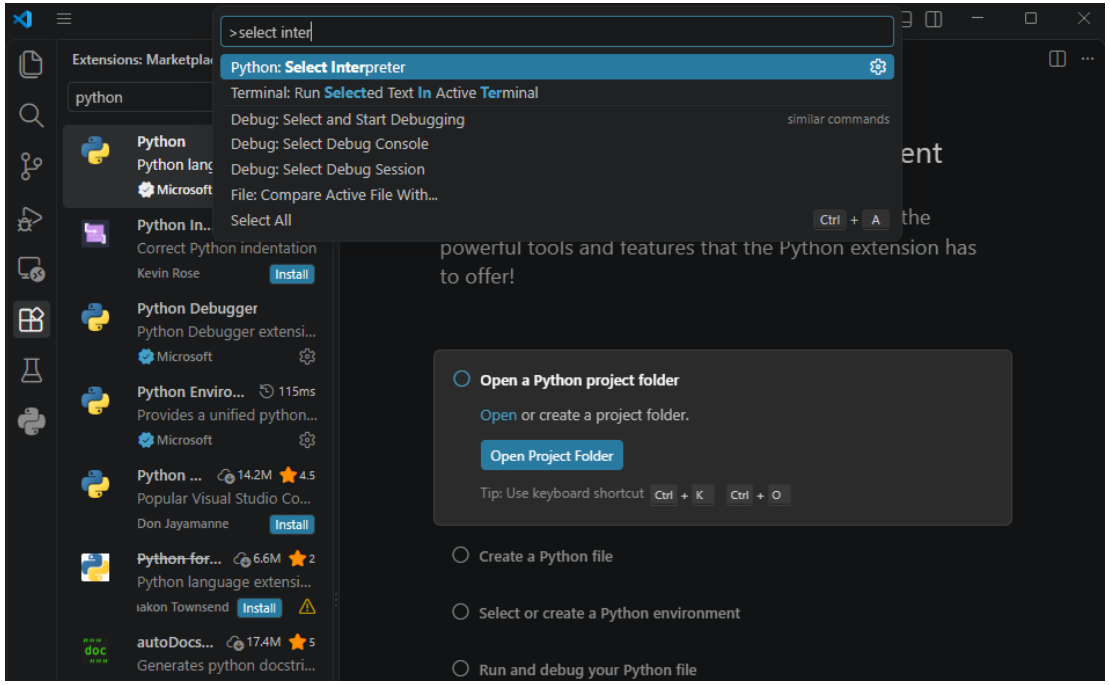
Step 3. Python Extension 설치하기

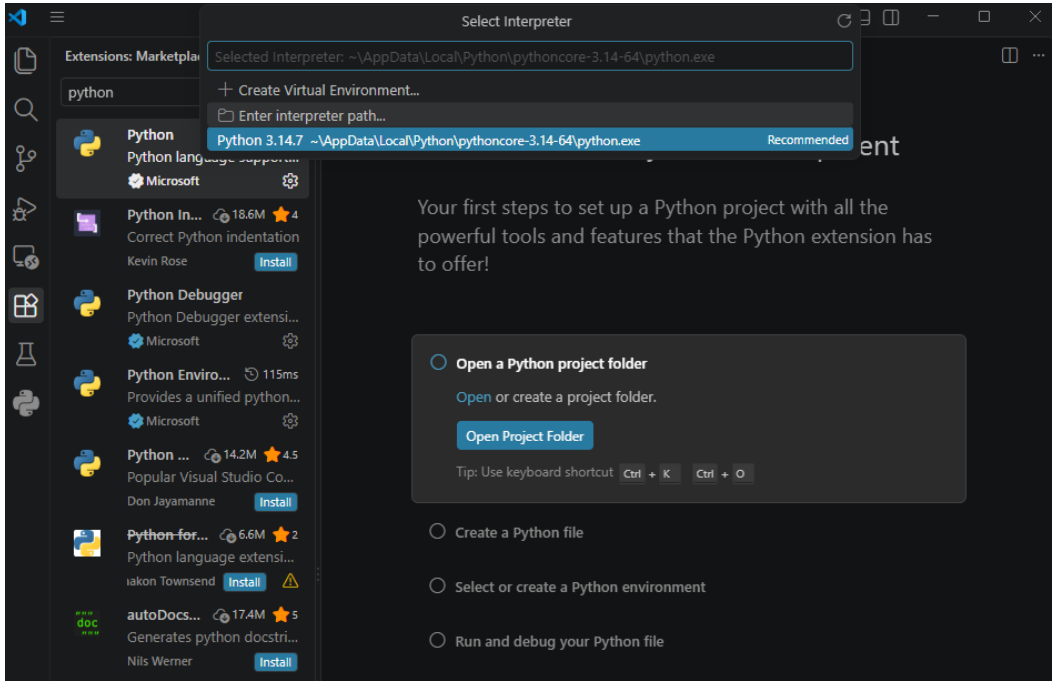


Python Extension은 VS Code에 Python 개발 기능을 추가한다.

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Step 4. Python Interpreter 선택하기



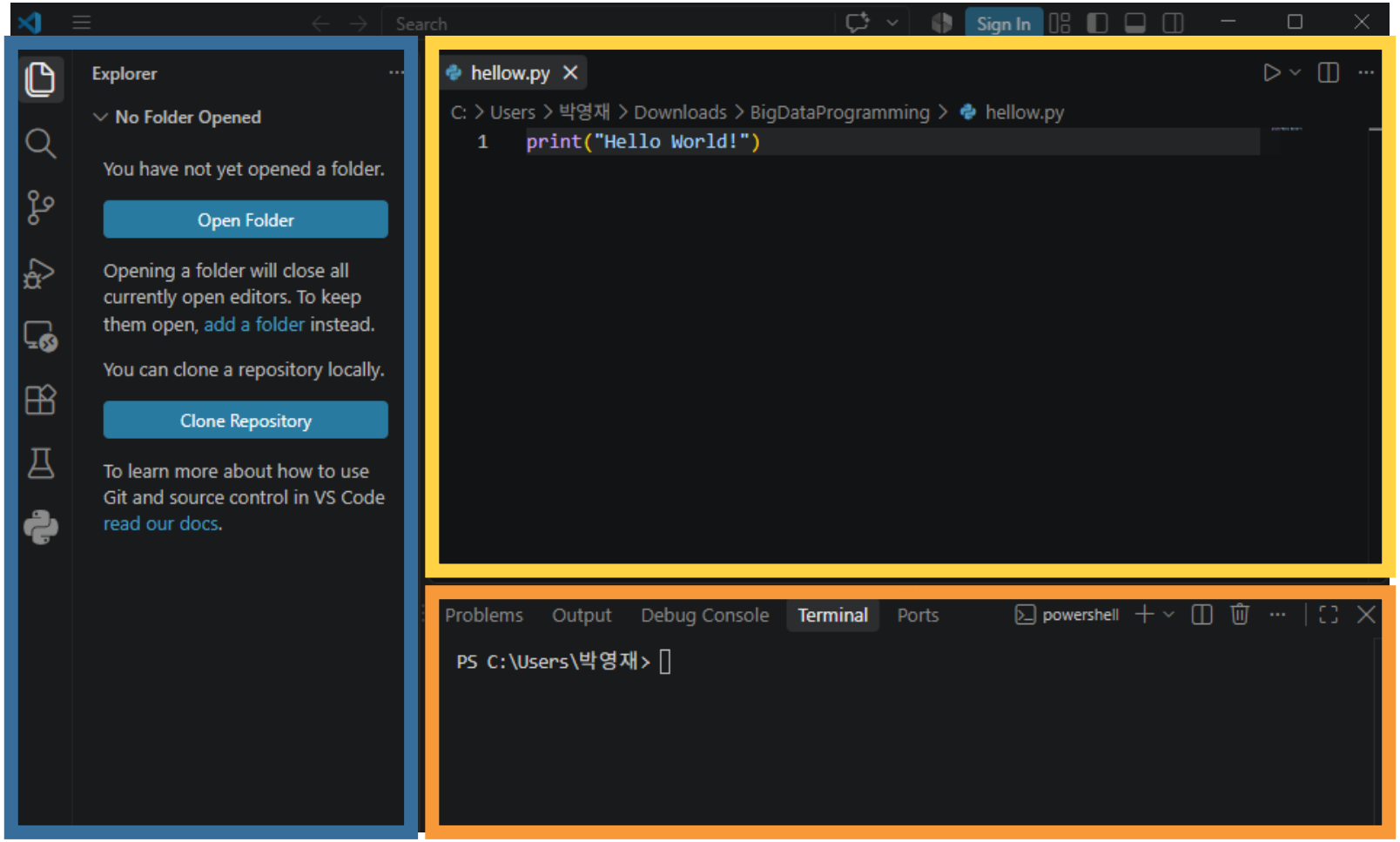


VS Code가 어떤 Python을 사용할 것인지 지정해야 한다.

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VS Code 화면 살펴보기

① Explorer



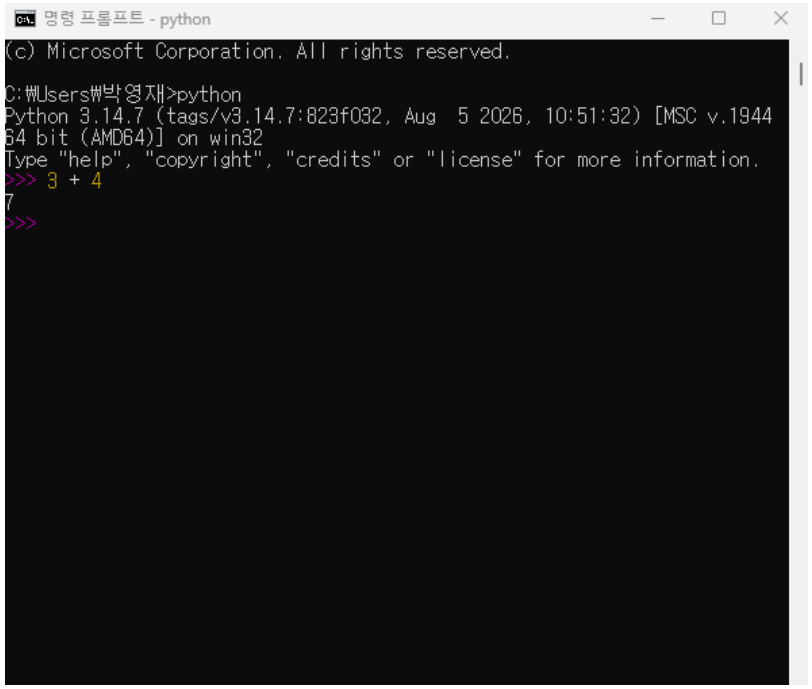
② Editor

③ Terminal

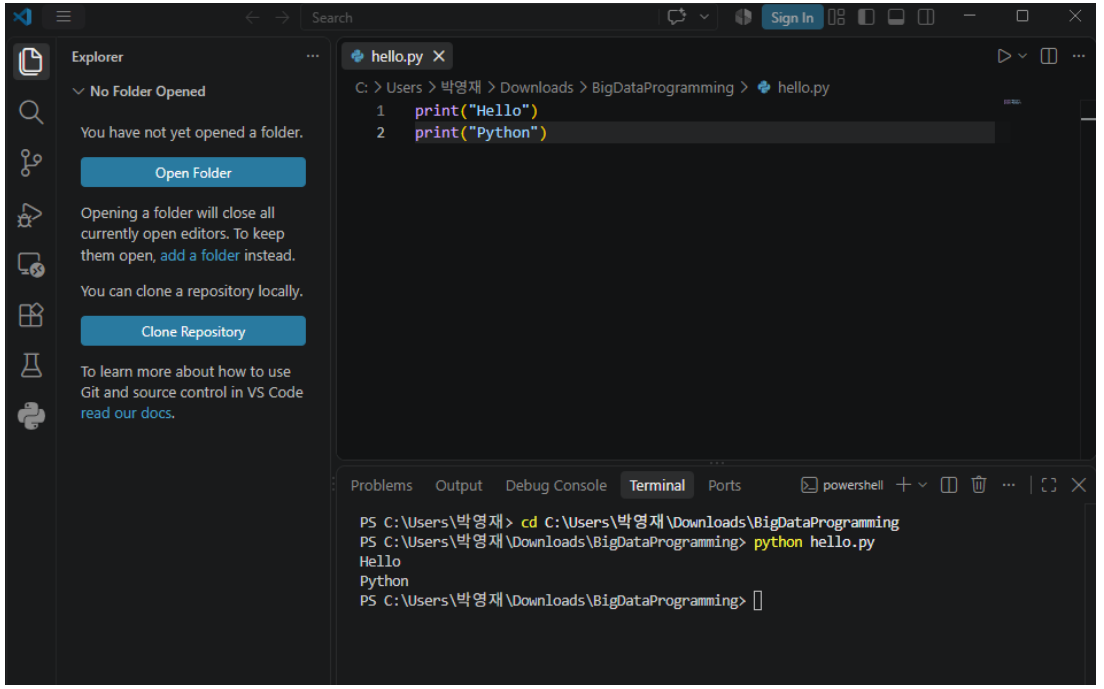
Explorer에서 파일을 관리하고, Editor에서 코드를 작성하며, Terminal에서 결과를 확인한다.

Running Python

Interactive Mode vs. Script Mode



Interactive Mode

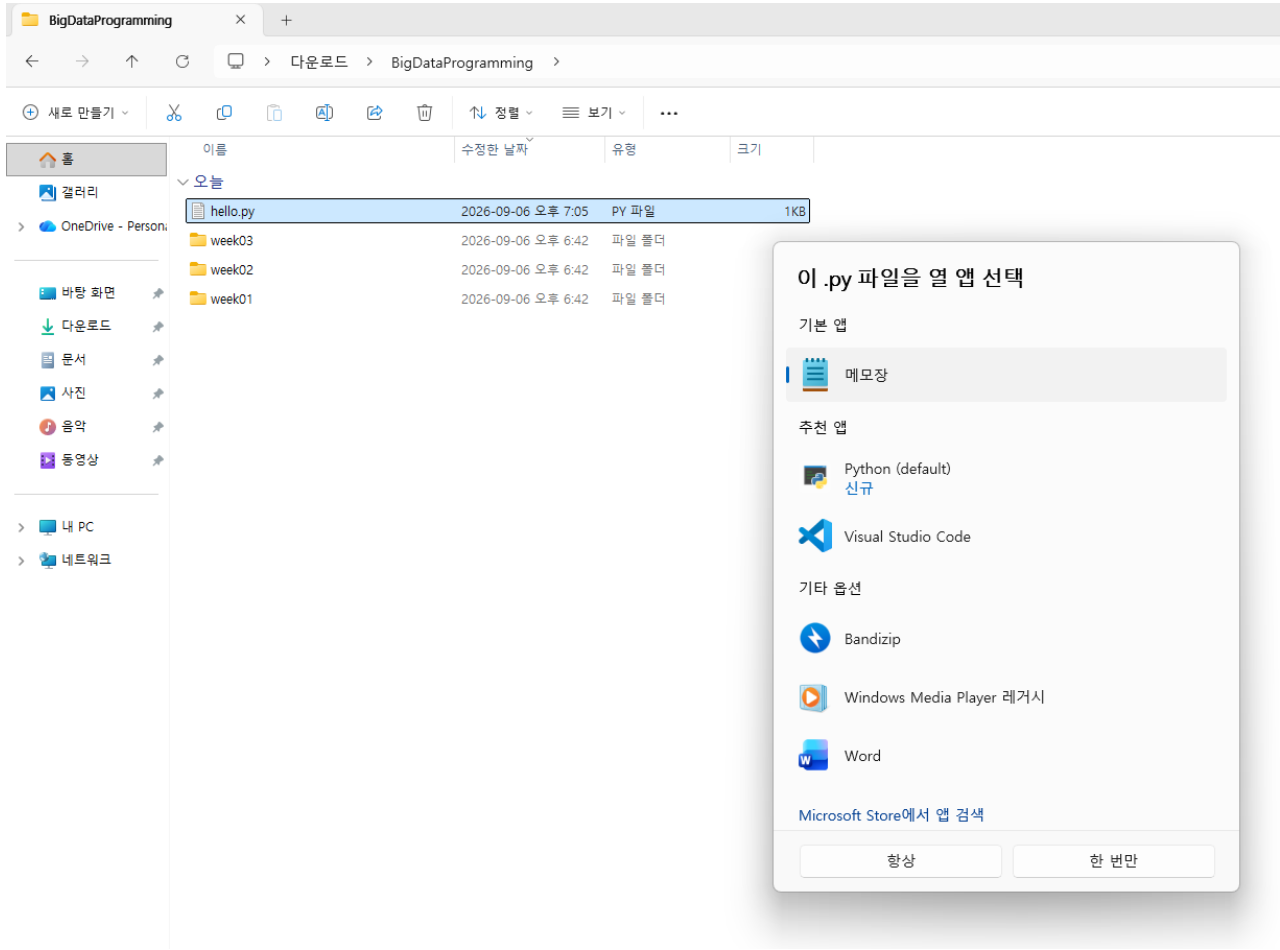


Script Mode

간단한 실험에는 Interactive Mode, 실제 프로그램 작성에는 Script Mode를 사용한다.

Running Python

Our First Python Program



Python 프로그램은 .py 확장자를 가진 소스 파일로 저장한다.